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Summary

Machine Learning researcher and **Imaging Science PhD candidate** specializing in **medical AI, computer vision, self-supervised representation learning, and 2D/3D medical image analysis**. Experienced in building **PyTorch and MONAI** workflows for real-world medical imaging data, including **CT and MRI preprocessing, segmentation, detection, classification, image enhancement, and model evaluation**. At Siemens Healthineers, developed **3D chest CT self-supervised pretraining** with a transformer backbone transferred to **organ classification, UNETR-based organ segmentation, and transformer-based lung nodule detection**. Published 7+ peer-reviewed papers at **ICML, NeurIPS, TMLR, and BMVC**, with research experience in representation learning, model generalization, error analysis, scalable medical AI experimentation, and a research interest in **multimodal medical AI**.

Technical Skills

- **Languages/Frameworks:** Python, PyTorch, MONAI, SimpleITK, Pydicom, OpenCV, Azure AI ML, Scikit-learn.
- **Deep Learning:** CNNs, Transformers, Self-Supervised Learning, Continual Learning, Representation Learning, Segmentation, Detection, Classification, Transfer Learning, Multi-GPU Training.
- **Medical AI:** 3D Chest CT, MRI, DICOM Processing, Medical Image Preprocessing, Resampling and Spacing, Intensity Normalization, CT Windowing, 3D Augmentation, Case-Level Evaluation.

Industry Experience

Siemens Healthineers 

Malvern, PA, USA 

ML Research Intern

August 2023 – November 2023

- Designed and implemented **3D self-supervised pretraining** for chest CT, producing a pretrained transformer backbone transferable across **organ classification, UNETR-based organ segmentation, and transformer-based 3D lung nodule detection**.
- Built reproducible **PyTorch/MONAI training and evaluation pipelines** for 3D chest CT data, including preprocessing, training, validation, and downstream evaluation on Azure AI ML.
- Improved downstream iteration efficiency (**up to 1000× fewer iterations**) for a transformer-based 3D lung nodule detection workflow through effective pretraining and representation transfer.
- *Technologies used:* Python, PyTorch, MONAI, Azure AI ML, UNETR, 3D chest CT data

NTT DATA Europe & LATAM 

Lima, Peru 

Data Scientist

May 2018 – July 2019

- Owned model training and evaluation for a **recommender system** that personalized promotions for **1.5M clients**; supported production deployment and communicated model performance to stakeholders.
- Built an **OpenCV video analytics pipeline** for indoor space usage estimation using object detection and tracking.
- Developed **CNN computer vision models** for applied visual classification tasks, including vehicle damage severity categorization and product recognition.
- *Technologies used:* Python, PyTorch, Keras, TensorFlow Lite, OpenCV, Docker, Flask, PySpark, DVC

Siemens Healthineers 

Malvern, PA, USA 

ML Research Intern

April 2017 – March 2018

- Designed and implemented a **3D CNN for lung nodule classification** on chest CT volumes, maintaining **90% sensitivity** while reducing **false positives per patient by 50%** compared with a production baseline.
- Built an annotation and case review tool to support expert label curation, iterative error analysis, and review of challenging chest CT cases.
- Presented model results, failure modes, and workflow usability findings to technical stakeholders.
- *Technologies used:* Python, Caffe, SimpleITK, Tkinter, 3D chest CT data

Universidad Nacional de Ingeniería 

Lima, Peru 

Research Assistant

July 2016 – March 2017

- Developed a **CNN medical image analysis model** for diabetic retinopathy detection from retinal fundus images, achieving 83% accuracy.
- Evaluated model performance and contributed to a publication on medical image classification.
- *Technologies used:* Python, Keras, OpenCV

IBM 

Lima, Peru 

Software Tester (QA)

April 2016 – October 2016

- Executed QA test strategies for a telecom database system; collaborated with engineers to reproduce issues, validate fixes, and improve software reliability.
- Built a prototype computer vision workflow using IBM Watson Visual Recognition services.
- *Technologies used:* Python, SQL, OpenCV, Watson Visual Recognition

Selected Project

MRI Fast Acquisition Enhancement *(Project Link)*

- Built a complete **MRI image quality enhancement pipeline** to restore knee MRI slices under simulated fast acquisition degradation using a 2D U-Net.
- Implemented DICOM-to-HDF5 conversion, per-volume normalization, degradation simulation, case-level splits, and evaluation on held-out test cases.
- Evaluated image quality with **PSNR, SSIM, and MAE**, supported by qualitative comparisons and error maps.
- *Technologies used:* Python, Pydicom, PyTorch, MONAI, PyTorch Lightning Fabric, DICOM, HDF5

Education

Rochester Institute of Technology 

Rochester, NY, USA 

Ph.D. Imaging Science

August 2019 – December 2026 (expected)

- **Research focus:** Self-supervised and continual representation learning for computer vision.
- **Relevant coursework:** Deep Learning for Vision, Human Visual System, Image Processing, Computer Vision
- **Highlights:** 7+ publications at ICML, NeurIPS, TMLR, BMVC, and other venues.

Universidad Nacional de Ingeniería 

Lima, Peru 

B.Sc. Mechatronics Engineering

March 2011 – December 2015

- **Relevant coursework:** Machine Learning, Image Processing, Object Oriented Programming
- **Highlights:** 2 publications

Selected Publications

Controlling Neural Collapse Enhances Out-of-Distribution Detection and Transfer Learning 

ICML 

Md.Y. Harun, J. Gallardo, and C. Kanan

2025

What Variables Affect Out-of-Distribution Generalization in Pretrained Models? 

NeurIPS 

Md.Y. Harun, K. Lee, J. Gallardo, G. Krishnan, and C. Kanan

2024

SIESTA: Efficient Online Continual Learning with Sleep 

TMLR 

Md.Y. Harun*, J. Gallardo*, T.L. Hayes, R. Kemker, and C. Kanan

2023

Self-Supervised Training Enhances Online Continual Learning 

BMVC 

J. Gallardo, T.L. Hayes, and C. Kanan

2021

Detection of Diabetic Retinopathy Based on a CNN Using Retinal Fundus Images 

ICANN 

G. García, J. Gallardo, A. Mauricio, J. López, and C. Del Carpio

2017

* *Equal contribution*

Research Service

Reviewer: WACV 2026; AAAI 2025–2026; ICCV 2025; CoLLAs 2025; CVPR 2024–2025; CLVISION 2024–2025